Features and tools, in depth

One page per feature and tool, each grounded in the code: what it does, every option, what it produces, how it works, a worked example, and the gotchas. These are the canonical reference. The usage guide is the short overview that links here; methods covers the underlying algorithms; output columns defines every result column.

Core analysis

- **Folding and temperatures**: MFE folding, the ViennaRNA model settings, and the 1-to-5 configurable temperatures.
- **Partition function**: ensemble energy, pairing probabilities, and RBS accessibility (the optional, slower path).
- **Hairpin detection**: terminal vs RBS-based detection and the stem-loop extraction.
- **RBS detection**: the configurable anchor and window, and RBS sequestering.
- **Composition**: full-sequence nucleotide frequencies and hairpin base-pair fractions.
- **Quality scoring**: criteria, weights, grace zones, tiers, and the Quality Score Builder.

Tools

- **Motif / Sequence Finder**: search every sequence for an IUPAC pattern and measure its sequestering.
- **Synthetic Pool Generator**: build random RNA pools from a segment template.
- **Sequence Extractor**: pull upstream/downstream regions from local files or NCBI.
- **RNArobo Search**: structural motif search (helices and loops).
- **Pseudoknot Finder**: pseudoknot prediction with Knotty.
- **RNAfold Deck**: tier hairpins → full-length and hairpin structures at 1–5 temperatures → verified PowerPoint, folded and drawn locally.